



Scan to view
live demo

INTELLIGENT WORKFORCE OPTIMIZATION SYSTEM

A Data-Driven Framework for Optimizing Technical Workforce Allocation
Femachi Ajoku & Peter Egessa · Intelligent Systems · Workforce Development · Adaptive Learning

2026
RISE

INNOVATION
SHOWCASE

SECTION 1 – PROBLEM STATEMENT

Modern project environments rely on **static assignment methods** that fail to account for the depth, adaptability, and real-world capability of technical talent.

Key Issues

- Skill matching is keyword-based, not competency-based
- Project allocation lacks dynamic adaptability
- Reduced productivity & higher failure rates
- Underutilization of skilled individuals

Industry Context

"Platforms like Jira and Asana primarily support task tracking — not intelligent talent allocation."

- No competency scoring in existing tools
- Gap between HR systems & project management

SECTION 3 – PROPOSED SOLUTION

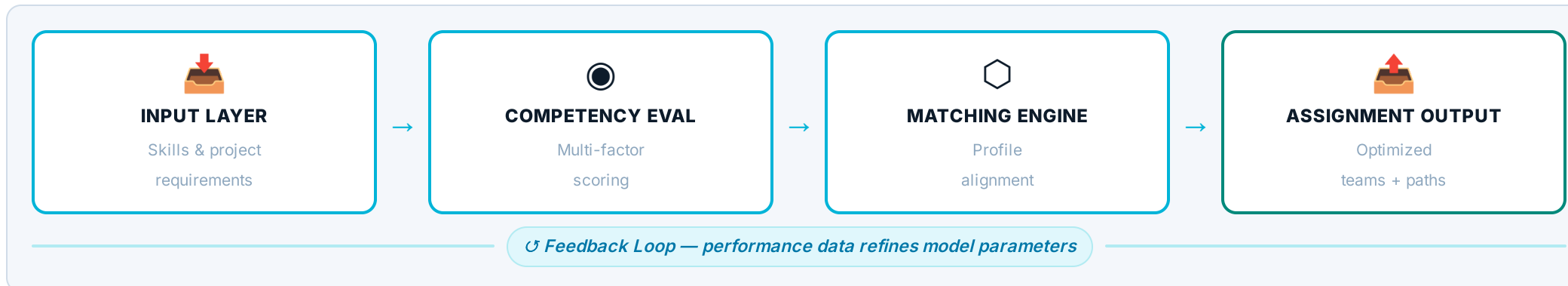
Concept

An intelligent allocation framework that analyzes technical capabilities and dynamically aligns individuals to projects.

Core Idea

Instead of static assignment, the system continuously evaluates competency profiles and aligns them with evolving project needs.

System Architecture



Module Descriptions

Input Layer Captures user skills and project requirements	Competency Evaluation Processes multi-factor capability indicators
Matching Engine Aligns individuals to tasks based on evaluated profiles	Feedback Loop Continuously improves alignment over time

SECTION 5 – INNOVATION

✗ TRADITIONAL SYSTEMS	✓ PROPOSED SYSTEM
✗ Static assignment	✓ Dynamic alignment
✗ Keyword matching	✓ Competency-based evaluation
✗ Manual decisions	✓ Data-driven framework
✗ Limited adaptability	✓ Continuous optimization

Key Innovations

- Intelligent interpretation of technical capability
- Dynamic responsiveness to project complexity
- Unified evaluation and assignment into one seamless system

SECTION 2 – OBJECTIVES

This project aims to:

- Develop a system that evaluates multi-dimensional technical competencies
- Enable adaptive matching between individuals and project requirements
- Improve efficiency, accuracy, and scalability in team formation
- Introduce a data-informed decision framework for project allocation

SECTION 4 – METHODOLOGY

Approach

The system applies a structured, multi-factor evaluation framework to assess alignment between individuals and project requirements.

Methodological Elements

- Multi-Dimensional Assessment**
Structured scoring across 6+ technical skill dimensions
- Pattern-Based Evaluation**
Evaluates skill relevance in real project context
- Adaptive Matching**
Aligns individuals based on contextual project needs
- Iterative Refinement**
Feedback integration improves accuracy each cycle

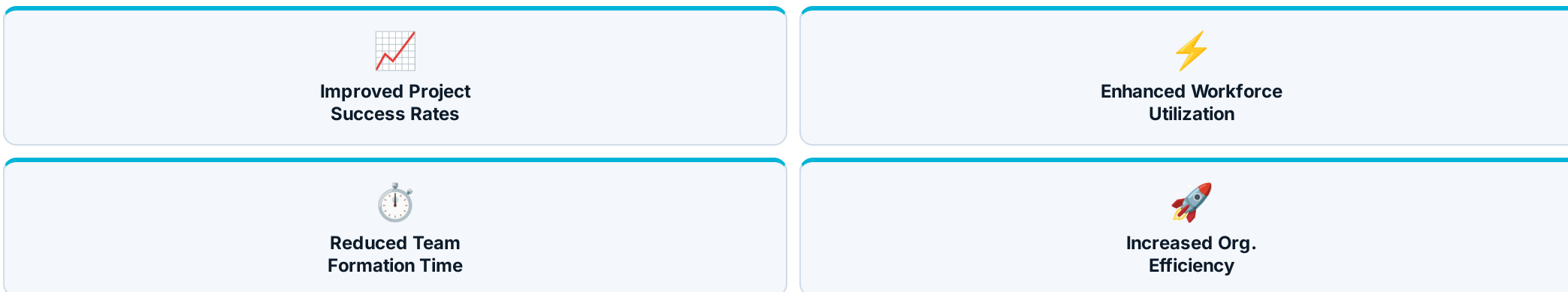
Rigor & Feasibility

- ✓ Scalable across teams and organizations
- ✓ Compatible with existing project management ecosystems
- ✓ Supports real-time decision-making environments

Certain technical components are proprietary and have been intentionally abstracted.

SECTION 6 – REAL-WORLD IMPACT

Expected Outcomes



Scalability

- Software engineering teams
- IT service environments
- Technical training platforms

Societal Relevance — UN Sustainable Development Goals

8 SDG Decent Work & Growth Productive employment & inclusive economic growth	9 SDG Industry & Innovation AI-driven methodology transforming talent deployment
4 SDG Quality Education Adaptive learning accelerates skill development	10 SDG Reduced Inequalities Merit-based matching removes bias

SECTION 7 – ENTREPRENEURIAL THINKING

Problem-Solving Approach

- Identifies inefficiencies in workforce allocation
- Applies structured analytical thinking to redesign assignment processes
- Bridges the gap between technical capability and project demand

Market Opportunity

No existing tool unifies **skill assessment + team formation + adaptive learning** into one intelligence loop. This is a new product category.

Growing Demand Intelligent workforce systems expanding across industries	Enterprise Integration Potential integration into Jira, Workday, SAP SuccessFactors	SaaS Deployment Scalable subscription tiers by team size & usage volume
--	---	---

SECTION 8 – FUTURE WORK

Next Steps:

- Refinement of proprietary evaluation models
- Expansion of system capabilities through real-world testing
- Integration with existing enterprise tools
- Performance optimization using live datasets

REFERENCES

- McKinsey & Co. (2023). AI in workforce optimization. mckinsey.com
- Harvard Business Review (2020). Reskilling in the age of automation. hbr.org
- Gartner (2023). Future of Work Trends Report. gartner.com
- United Nations (2015). Sustainable Development Goals. un.org/sdgs
- IEEE (2023). Adaptive learning systems in enterprise environments.
- Teneo AI (2024). Intelligent workforce matching systems. teneo.ai
- ResearchGate (2022). Competency-based talent management.
- Fuel Innovation Orilla (2025). Bright Minds Challenge One.
- Georgian College RISE (2026). Project submission guidelines.
- Roboflow (n.d.). Computer vision capability tools. roboflow.com